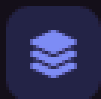


BuildGram

Building an Instagram-inspired platform end-to-end.
Master modern web development through real-world challenges.



CORE STACK

React • Python / Golang •
PostgreSQL



COMMITMENT

8 Weeks • 7–10 hrs/week
Moderate to
Advanced



PROJECT OUTCOME

Production-minded app with clean APIs
& data consistency

Project Overview

Objective

Guide mentees to build a feature-complete social media app end-to-end, mimicking real-world development workflows.

Scope

- Comprehensive user accounts, creating posts with images, and interactive likes and follows.
- Intelligent, personalized feed generation with logic-based content ranking.

Technical Focus

- **Frontend:** React (component-based UI, efficient state management, and declarative routing).
- **Backend:** REST APIs in Python or Golang (structured routing, controllers, middleware, JWT).
- **Database:** PostgreSQL (relational schema design, multi-table relationships, indexing, optimization).
- **Cross-cutting:** Robust API design, data consistency across transactions, and performance tuning.

Learning Outcomes & Tech Stack

Learning Outcomes

- Design component-based UIs and manage application state efficiently.
- Build secure REST APIs with robust authentication and middleware layers.
- Model relational data, write optimized SQL queries, and integrate databases seamlessly.
- Implement complex features like likes/follows and logic-driven feed ranking.
- Plan, version, and document APIs while collaborating effectively in a team environment.
- Deliver a fully working demo with clean, maintainable code structure.

Tech Stack

- **Frontend**
React, React Router, Fetch/Axios
- **Backend**
Python or Golang
- **Database**
PostgreSQL
- **Auth**
JWT, bcrypt
- **Tooling**
Git/GitHub, Postman, Docker (optional)

Week-wise Schedule (8 Weeks)

Week 1-2 : Frontend Fundamentals

- **React Basics:** Components, props, state management, and declarative routing.
- **Focus:** Building a clean component structure and separating UI logic from rendering.

Week 3-4 : Backend Fundamentals

- **REST APIs:** Developing structured APIs using Golang or Node.js.
- **Architecture:** Implementing routing, controllers, and middleware separation.

Week 4-5 : Database Design + Integration

- **PostgreSQL:** Designing tables, multi-table relationships, and constraints.
- **Integration:** Connecting the backend to the database and writing optimized core queries.

Week 5-6 : Auth + Relations

- **Security:** Implementing JWT authentication, authorization, and password hashing.
- **Domain Logic:** Building likes/follows systems, ensuring data integrity, and completing the mid-term checkpoint.

Evaluation & Assignment Outline

Assessment Methods

- **Weekly Assignments (4):** 40%
- **Mid-Term Check:** 20% (Week 4 demo + review)
- **Final Group Project:** 35% (App)
- **Professionalism:** 5% (Syncs, PR hygiene)

Assessment Rubric

- Correctness and completeness of features
- Code quality, modularity, and structure
- API design and documentation clarity
- DB schema design and query performance
- Security best practices (JWT, hashing)
- UX polish and interface responsiveness

Deliverables & Formats

- **Weekly Requirements**
Assignment Submission on GitHub with README.
- **Mid-Term Evaluation**
Running application (local/cloud), schema diagrams, and generated API docs.
- **End-Term Evaluation**
Group repository, deployed platform demo.

Commitment & Prerequisites

Requirements

- **Commitment**

7–10 hrs/week (coding, learning, sessions)

- **Difficulty Level**

Moderate to Advanced (adaptive based on progress)

- **Expectations**

Consistent weekly commits; attend sessions

- **Prerequisite**

Enthu and willingness to learn

- **Prior Knowledge**

Git/GitHub basics

Mentors & Contact Details



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